BioMedical Natural Language Agents: A Search and Generation Pipeline for TREC BioGen

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Abstract.
- **Task:** TREC BioGen 2024 — biomedical question answering with citation grounding over 4,194 PubMed abstracts, 65 topics.
- **Phase 1 (IR pipeline):** five retrieval strategies (BM25, LM-Jelinek-Mercer, LM-Dirichlet, Dense KNN, RRF); disciplined hyperparameter tuning via 5-fold CV on 32 train queries; field ablation, encoder comparison, statistical significance tests.
- **Best result:** KNN(MedCPT) — NDCG@100=0.8077, MAP=0.6143, Recall@100=0.8933 on 33 test topics (Phase 2 locked strategy); RRF (BM25+MedCPT KNN, k=60) reaches NDCG@100=0.8245 on the same set; both statistically significant vs. BM25 baseline (p=0.0004).
- **Key finding:** domain-specialised dense encoding (MedCPT, trained on PubMed citation graph) is the single most impactful component (ΔNDCG@100=+0.0501 vs. general-purpose baseline).
- **Phases 2–3 (planned):** cross-encoder reranking and LLM-augmented generation (vLLM, MedCPT-Cross-Encoder, LLM-as-judge); then a ReAct agentic loop for multi-step biomedical synthesis.

1 Introduction

- LLMs have significantly advanced biomedical NLP tasks: question answering, literature summarisation, clinical note processing.
- Key challenge: **hallucinations and confabulations** — LLMs generate plausible but unsupported claims, especially dangerous in clinical settings.
- Grounding LLM outputs in retrieved evidence is the standard mitigation strategy (Retrieval-Augmented Generation, RAG).
- TREC BioGen 2024/2025 [2] formalises this as a shared task: retrieve PubMed abstracts, generate a grounded answer, and cite supporting evidence.
- **Task definition:** given a biomedical question (topic + clinical question + narrative), retrieve relevant PubMed abstracts and eventually generate a paragraph-length answer with sentence-level citations.
- **Why this is hard:**
  - Biomedical vocabulary gap: clinical queries use lay or mixed terminology; PubMed abstracts use precise scientific notation.
  - Precision vs. recall tension: missing a supporting study is as harmful as citing a contradicting one.

  - Citation-level granularity: the answer must ground every claim, not just list references.
  - Evaluation complexity: requires IR metrics (MAP, NDCG) *and* generation quality metrics (citation accuracy, factual consistency).

- **Phase 1 — Search & Evaluation (this report):**
  - Index 4,194 PubMed abstracts in OpenSearch.
  - Implement and tune 5 retrieval strategies: BM25, LM Jelinek-Mercer, LM Dirichlet, dense KNN, and Reciprocal Rank Fusion (RRF).
  - Evaluate with MAP, MRR, P@10, R@100, NDCG@100 against citation-derived relevance judgements.
  - Best Phase 1 result: **KNN (MedCPT) NDCG@100 = 0.8077, MAP = 0.6143, R@100 = 0.8933** on 33 held-out test queries. RRF (BM25+MedCPT, k=60) reaches NDCG@100=0.8245.
- **Phase 2 — RAG & LLM Judges (upcoming):**
  - Cross-encoder reranking of retrieved evidence.
  - LLM-generated paragraph answers with per-sentence citations.
  - LLM-as-a-judge evaluation of answer quality and citation faithfulness.
- **Phase 3 — Deep Research Agent (upcoming):**
  - ReAct-style agentic loop: plan sub-topics, retrieve per sub-topic, aggregate evidence, write structured report.
  - Multi-step reasoning over biomedical literature.

- **Contributions (Phase 1):**
  - Full IR pipeline from raw JSONL corpus to evaluated ranked lists, implemented from scratch following TREC and Lab03 protocols.
  - Systematic hyperparameter tuning with strict train/test discipline (5-fold CV on 32 train queries; test set touched once).
  - Encoder comparison demonstrating that domain-matched MedCPT yields +0.0501 NDCG@100 over general-domain msmarco-distilbert on the test set (0.7576→0.8077, +42.8% relative on train CV).
  - Post-hoc analyses: statistical significance tests, IDF-based difficulty characterisation, Jaccard overlap, confusion matrix at P@10, query/document length correlations.

This report is organised as follows. Section 2 describes the BioGen NL Agent pipeline across three phases. Section 3.1 details the experimental setup (datasets, metrics, protocols). Section 3.2 presents Phase 1 results and discussion. Section 4 concludes and outlines future work.

2 BioGen NL Agent

2.1 Data Parsing, Indexing and Search (Phase 1)

Corpus

- **Source:** `filtered_pubmed_abstracts.txt` — JSONL, one document per line; fields: `id` (PMID string), `contents` (title + abstract concatenated).

- **Size:** 4,194 PubMed abstracts; parsed via `json.loads()` per line (not `json.load()` — file is not a JSON array).
- **Document length:** mean=150 words, median=142, min=12, max=964. High variance ($\sigma \approx 100$ words) motivates length normalisation in BM25 and LM-Dirichlet.
- **Indexing unit:** concatenated title+abstract as-is; no splitting, no preprocessing beyond lowercase tokenisation at query time.

Topics

- **Source:** `BioGen2024topics.json` — 65 TREC BioGen 2024 topic entries, IDs 116–180.
- **Fields per topic:** `topic` (short keyword, 2–6 words), `question` (natural-language clinical question, 10–20 words), `narrative` (retrieval context, 30–70 words).
- **Query field ablation:** three formulations evaluated — `topic`-only, `question`-only, and `topic+question` concatenation.
  - `question` and `topic+question` are top performers; `topic+question` locked as the final field (NDCG@100=0.7833 vs 0.7174 for `question`).
  - `narrative` alone is weakest (NDCG@100 = 0.6099 vs 0.7833 for `topic+question`).

Train/Test Split

- **Protocol:** fixed odd/even split on topic IDs — a professor-mandated requirement for the course.
  - **Train:** 32 queries (odd IDs: 117, 119, . . . , 179).
  - **Test:** 33 queries (even IDs: 116, 118, . . . , 180).
- **Corpus not split:** all 4,194 documents are shared between train and test — only the query set is partitioned.
- **Train set used for:** query field selection, LM-JM λ sweep, BM25 k_1/b sweep, LM-Dir μ sweep, encoder comparison (5-fold CV).
- **Test set:** touched only once — final evaluation after all parameters are locked.
- **Saved to:** `results/splits/train_queries.json`, `results/splits/test_queries.json`.

Relevance Judgements (qrels)

- **Source:** `biogen_2024_submissions.json` — automated system submissions from BioGen 2024 with sentence-level citation assessments (`evidence_relation` field).
- **Evidence relation value counts:** supporting=10,240; neutral=1,412; not relevant=2,201; contradicting=275; invalid citation=635.
- **Binary qrels** (for MAP, MRR, P@k, R@k):
  - Rule: `evidence_relation == "supporting"` ⇒ score=1; all others ⇒ score=0.
  - 65/65 topics have ≥1 relevant document.

  - Mean relevant docs/topic = 46.1 (range: 8–88).
  - Total relevant pairs: 2,999 (topic, PMID) pairs across 65 topics.
- **Graded qrels** (for NDCG@100):
  - Rule: supporting→5, neutral→2, all others→0. Per (topic, PMID) pair: take the maximum score across all citations.
  - Score distribution: 2,999 pairs at score=5; 218 pairs at score=2.
  - Total graded pairs: 3,217; mean relevant docs/topic = 49.5.
- **Quality note:** citation-based from automated systems, not human assessors. Mean 46.1 relevant docs/topic is far above standard TREC (3–15). MAP/recall denominators are inflated; not directly comparable to public TREC benchmarks.
- **Out-of-corpus PMIDs:** PMID 37711029 (topic 141) absent from the filtered corpus — excluded with a logged warning. 11 additional PMIDs in neutral-only citations are also excluded.

Index Design

- **Server:** OpenSearch at `api.novasearch.org:443`, shared course instance; index name = `usernlp03`.
- **One index, multiple similarity fields** — not one index per strategy. Shards=4, replicas=0, `refresh_interval=-1` during indexing.
- **Three custom similarity plugins registered at index creation:**
  - `lmjm_01`: LMJelinekMercer λ=0.1
  - `lmjm_07`: LMJelinekMercer λ=0.7 (baseline field)
  - `lmdir`: LMDirichlet μ=2,000 (default baseline)
- **Six index fields:**
  - `doc_id` — keyword (PMID, exact match)
  - `contents` — text, standard analyser, BM25 similarity
  - `contents_lmjm_01` — text, standard, LMJelinekMercer λ=0.1
  - `contents_lmjm_07` — text, standard, LMJelinekMercer λ=0.7 (baseline)
  - `contents_lmjm_065` — text, standard, LMJelinekMercer λ=0.65 (tuning winner)
  - `contents_lmdir` — text, standard, LMDirichlet μ=2,000
  - `embedding` — knn_vector, dim=768, HNSW (inner product, `ef_construction`=256, M=48)
- **Analyser choice:** `standard` (tokenise + lowercase) over `english` (Porter stemming) — stemming mangles biomedical terms (e.g. *hepatitis* → *hepat*).
- **KNN similarity:** `innerproduct` with L2-normalised vectors ⇒ inner product ≡ cosine similarity; no per-query renormalisation overhead.
- **Tuning sweep fields added later:** e.g. `contents_lmdir_75`, `contents_bm25_k15_b10` — created by the tuning pipeline without modifying the base index structure.
- **Idempotent creation:** `create_index()` checks `indices.exists()` first — never deletes or overwrites.

Dense Encoder

- **Baseline:** `msmarco-distilbert-base-v2` — DistilBERT fine-tuned on 8.8M MS MARCO (web) passage pairs; 768-dim output.
- **Encoding:** mean pooling over all token embeddings, L2-normalised via `F.normalize(p=2, dim=1)`.
- **Corpus encoding:** 4,194 docs in ≈3 min on CPU (batch=32); saved to `results/embeddings/msmarco.npy` (12.3 MB).
- **Improved encoder (tuning):** `ncbi/MedCPT-Query-Encoder` + `ncbi/MedCPT-Article-Encoder` [3] — asymmetric, trained on 23M PubMed search click-through pairs.
  - Exact domain match: PubMed query → abstract, same as our task.
  - Train CV NDCG@100: +0.0837 over msmarco (0.7448→0.8273).
  - Test NDCG@100 improvement: +0.0501 (0.7576→0.8077).
  - New index field `embedding_medcpt` (768-dim) added without rebuilding the base index.
- **multi-qa-mpnet-base-cos-v1** also evaluated: train CV NDCG@100=0.8236 vs MedCPT=0.8273 (gap of 0.0037). multi-qa wins MAP (0.6435) and P@10 (0.7719) on train but loses NDCG@100 — MedCPT locked as primary encoder.

Retrieval Strategies

- **Top-k = 100 documents** per query for all strategies.
- **Uniform interface:** all retrievers implement `BaseRetriever.search(query, size=100)` → `list[(pmid, score)]`.
- **BM25** — `match` on `contents` field [4]; default: k_1=1.2, b=0.75. Tuned: k_1=1.5, b=1.0.
  - b=1.0 (full length normalisation) outperforms b=0.75 across all k_1 values — abstracts vary widely (12–964 words).
  - k_1 change has minimal impact; term saturation is not the bottleneck for biomedical multi-term queries.
- **LM Jelinek-Mercer** — `match` on `contents_lmjm_065` (λ=0.65, extended grid winner; ΔNDCG=+0.001 over original λ=0.7).
  - λ=0.65 gives more weight to the document's own term distribution — correct for 10–20 word clinical queries.
  - λ=0.9 is nearly tied (ΔNDCG=0.0001, within noise); λ=0.65 locked (extended grid winner).
- **LM Dirichlet** — `match` on `contents_lmdir_100` (μ=100, CV winner; baseline used μ=2,000).
  - Default μ=2,000 ≈13× mean doc length (150 words) — over-smooths every abstract.
  - μ=100 is the CV winner (NDCG@100=0.7706); Δ=+0.0196 vs default μ=2000 (0.7510).
  - Sweep: NDCG@100 peaks at μ=100, monotone decrease toward μ=2,000.
- **Dense KNN** — HNSW approximate nearest-neighbour on `embedding_medcpt` field; query encoded with MedCPT Query-Encoder.

  - Cosine similarity via inner product of L2-normalised vectors.
  - `ef_search`=100 at index level; brute-force exact cosine used during encoder comparison (no HNSW approximation error).
- **RRF Fusion** [1] — Reciprocal Rank Fusion of BM25 + KNN (MedCPT), k=60 (Cormack default; locked for Phase 2 backbone).
  - Score: $\mathrm{RRF}(d) = \sum_r \frac{1}{k+\mathrm{rank}_r(d)}$.
  - BM25 and KNN(MedCPT) Jaccard overlap at top-10: 0.13–0.16 — very low, confirming complementary signals.
  - RRF pair sweep (5-fold CV): best lexical pair was BM25+LM-Dir k=90 (NDCG@100=0.7907); no RRF pair beats KNN(MedCPT) solo (0.8273) on the train CV leaderboard.
  - Test-set RRF (BM25+MedCPT, k=60) achieves NDCG@100=0.8245 — the highest test-set score overall, with strong R@100=0.8924. Best solo model is KNN(MedCPT) with NDCG@100=0.8077.

Implementation Notes

- **Source layout:** `src/data/`, `src/indexing/`, `src/retrieval/`, `src/embeddings/`, `src/evaluation/`, `src/tuning/`.
- **Run files:** `results/phase1/{strategy}_run.json` — format `{topic_id: [(pmid, score), ...]}`; one file per strategy, baseline and tuned variants.
- **Bulk indexing:** 4,194 docs in ≈21 s using `opensearchpy.helpers.bulk()`, batch=100; idempotent count-check before re-indexing.
- **Colab compatibility:** all notebooks detect `IN_COLAB` and adapt paths and credential loading (Colab Secrets vs. local `.env`).

2.2 LLM Augmented Generation (Phase 2)

[PLACEHOLDER — Phase 2 not yet implemented.] Results, figures, and final analysis will replace this section after Phase 2 completion. The bullet points below document the planned design.

- **Goal:** generate a grounded paragraph-length answer per topic, citing specific PubMed abstracts as evidence, using the Phase 1 retrieval backbone as the evidence source.
- **Retrieval backbone (inherited from Phase 1):**
  - Phase 2 locked strategy: KNN(MedCPT), with RRF fallback (BM25 k_1=1.5, b=1.0 + MedCPT KNN, k=60).
  - KNN(MedCPT): NDCG@100=0.8077, R@100=0.8933; RRF: NDCG@100=0.8245, R@100=0.8924 — both retrieve >89% of relevant docs in top 100, providing a rich evidence pool for generation.
- **Cross-encoder reranking (planned):**
  - Model: `ncbi/MedCPT-Cross-Encoder` or BioBERT-based cross-encoder.
  - Input: (query, candidate abstract) pairs; output: relevance score.
  - Select top-3 supporting sentences per abstract for citation injection.
  - Reranked list passed to the generation step.

- **LLM generation (planned):**
  - Server: vLLM at `amalia.novasearch.org` via OpenAI-compatible API.
  - Prompt design: system role + retrieved abstracts (top-k) + question → paragraph answer with inline citations.
  - Constraints (TREC BioGen 2025): ≤2,500 words, ≤3 PMIDs per sentence, only PMIDs from the corpus.
  - Citation injection: post-process raw LLM output to add bracketed PMID references after each sentence.
- **LLM-as-judge evaluation (planned):**
  - GPT-4o via IAedu API as judge model (not vLLM).
  - Sentence-level alignment: judge classifies each generated sentence as supported / contradicted / ungrounded by cited abstracts.
  - Metrics to report: citation accuracy rate, factual consistency, hallucination rate.
- **Embedding and attention analysis exercise (planned):**
  - Positional embeddings: visualise distance from token 1 across 200 repeated tokens (2D scatter, distance matrix).
  - Contextual embeddings: visualise layer 0 to layer 11 evolution.
  - Self-attention: examine cross-encoder attention patterns for query-evidence alignment.
- *[To be completed after Phase 2 implementation. All results and analysis will be added here.]*

2.3 LLM Agentic Patterns (Phase 3)

[PLACEHOLDER — Phase 3 not yet implemented.] Results, figures, and final analysis will replace this section after Phase 3 completion. The bullet points below document the planned design.

- **Goal:** build a Deep Research Agent that autonomously plans sub-topics, retrieves evidence per sub-topic, aggregates findings, and writes a structured biomedical report with grounded citations.
- **Architecture (planned) — ReAct-style agentic loop:**
  - **Planner:** topic → list of sub-topics via LLM (*"What are the main dimensions of this biomedical question?"*).
  - **Explorer:** for each sub-topic, run the Phase 1+2 pipeline (retrieve, rerank, extract supporting sentences).
  - **Aggregator:** merge evidence across sub-topics; deduplicate citations; verify consistency across sub-claims.
  - **Report writer:** compose structured final report with section headings, per-claim citations, and a summary.
- **Key design decisions (planned):**
  - Agent calls `retrieval.search(sub_topic)` and `generation.generate(docs, sub_topic)` from shared `src/` — no code duplication across phases.
  - Each ReAct iteration: Reason → `SEARCH` → Observe → ... → Generate.
  - Stopping criterion: fixed max iterations or LLM confidence threshold.

- **Evaluation (planned):**
  - Citation accuracy: fraction of agent-cited PMIDs that are in the qrels (binary precision).
  - Sub-topic coverage: do the planned sub-topics collectively address the original clinical question?
  - Comparison against Phase 2 single-shot generation: does multi-step planning improve factual coverage?
- *[To be completed after Phase 3 implementation. All results and analysis will be added here.]*

3 Evaluation

3.1 Experimental Setup: Datasets, Metrics and Protocols

Dataset

- **Collection:** TREC BioGen 2024 — 4,194 PubMed abstracts, 65 biomedical topics (IDs 116–180).
- **Topic fields:** `topic` (keyword), `question` (natural language), `narrative` (retrieval context).
- **Relevance source:** citation assessments from automated BioGen 2024 system submissions — *not* human-assessed.
- **Binary qrels:** supporting=1, all others=0. Mean 46.1 relevant docs/topic (range 8–88).
- **Graded qrels:** supporting=5, neutral=2, others=0. Mean 49.5 relevant docs/topic.
- **Important caveat:** the high mean relevant-doc count (46.1) is 3–15× above standard TREC tasks. MAP and recall values are not comparable to published IR benchmarks.

Split Protocol

- **Fixed odd/even split** on topic IDs (professor-mandated).
- **Train:** 32 queries (odd IDs) — hyperparameter tuning only.
- **Test:** 33 queries (even IDs) — final evaluation, touched once.
- **Corpus:** shared, never split.
- **Hyperparameter selection protocol:**
  1. Query field ablation (BM25 on train) → lock field.
  2. LM-JM λ sweep on train (5-fold CV) → lock λ.
  3. BM25 k_1/b sweep on train (5-fold CV) → lock k_1, b.
  4. LM-Dir μ sweep on train (5-fold CV) → lock μ.
  5. Dense encoder comparison on train (exact cosine) → lock encoder.
  6. Run all strategies on test set once — no further tuning.
- **5-fold cross-validation** on 32 train queries: ≈26 queries train / 6 queries validation per fold. Primary criterion: mean NDCG@100 across folds.

Metrics

- All metrics implemented from scratch in `src/evaluation/metrics.py` (plain Python + NumPy; no external IR library — Lab03 protocol).
- TREC standard: 0-relevant topics excluded from MAP and MRR averages (none in this dataset — all 65 topics have ≥1 relevant doc).
- **MAP** (Mean Average Precision, binary qrels):
  - $\mathrm{AP} = \frac{1}{|\mathcal{R}|} \sum_{k=1}^{n} P(k) \cdot \mathrm{rel}(k)$, where $\mathcal{R}$ = set of relevant docs.
  - Measures overall ranking quality across all 100 retrieved docs.
  - Sensitive to how many relevant docs are found in top-100.
- **NDCG@100** (Normalised Discounted Cumulative Gain, graded qrels) — **primary TREC BioGen metric**:
  - $\mathrm{DCG} = \sum_{i=1}^{100} \frac{2^{\mathrm{rel}_i}-1}{\log_2(i+2)}$; $\mathrm{NDCG} = \mathrm{DCG}/\mathrm{IDCG}$.
  - Uses graded relevance (supporting=5, neutral=2).
  - Discounts documents ranked lower — rewards finding supporting evidence early.
- **MRR** (Mean Reciprocal Rank, binary qrels):
  - $\mathrm{MRR} = \frac{1}{|Q|} \sum_{q} \frac{1}{\mathrm{rank}_q^{(1)}}$.
  - Measures how quickly the first relevant document appears.
- **P@10** — precision in the top 10 results (binary qrels). Most relevant for clinical use: what a clinician sees on screen.
- **R@100** — recall at rank 100 (binary qrels). Critical for Phase 2: fraction of all relevant docs available as evidence.
- **PR Curves** — 11-point interpolated precision-recall curves, averaged across all test queries (Lab03 pattern).
- **Per-topic AP distribution** — box plots of AP per strategy to assess consistency across topics.

Locked Configuration (Phase 1 Final)

- **Query field:** `topic+question` (topic keyword concatenated with clinical question).
- **BM25:** k_1=1.5, b=1.0 — CV NDCG@100=0.7989.
- **LM Jelinek-Mercer:** λ=0.65 (extended grid winner) — CV NDCG@100=0.7754. Original grid 0.1,0.3,0.5,0.7,0.9 found λ=0.7 (0.7746); adding λ=0.65 and 0.75 confirmed 0.65 as the optimum (Δ=+0.0008, within noise).
- **LM Dirichlet:** μ=100 (CV winner) — CV NDCG@100=0.7706 (Δ=+0.0196 vs default μ=2000).
- **Dense encoder:** MedCPT (asymmetric) — CV NDCG@100=0.8273.
- **Fusion (Phase 2 backbone):** RRF k=60, tuned BM25 + MedCPT KNN — test NDCG@100=**0.8245**.
- **Best overall on train CV:** KNN(MedCPT) NDCG@100=0.8273. Best overall on test set: RRF (BM25+MedCPT) NDCG@100=**0.8245**.

3.2 Results and Discussion

Query Field Ablation (BM25, Train Set)

- Tested on 32 train queries; BM25 default parameters as proxy retriever.
- **Winner:** `topic+question` — NDCG@100=0.7833.
- `narrative` adds noise (instructions, qualifiers not in abstracts); `topic` alone misses clinical context synonyms.

Table 1. Query field ablation — BM25 default on 32 train topics. Bold = winner; NDCG@100 is the selection criterion.

Field	MAP	MRR	P@10	NDCG@100
topic only	0.5334	0.8255	0.6313	0.7381
question only	0.4988	0.8135	0.6562	0.7174
narrative only	0.3900	0.6890	0.5500	0.6099
topic+question	**0.5843**	0.8229	**0.7000**	**0.7833**
topic+narrative	0.5145	0.7083	0.6500	0.7205
all three	0.5673	**0.7240**	0.7031	0.7611

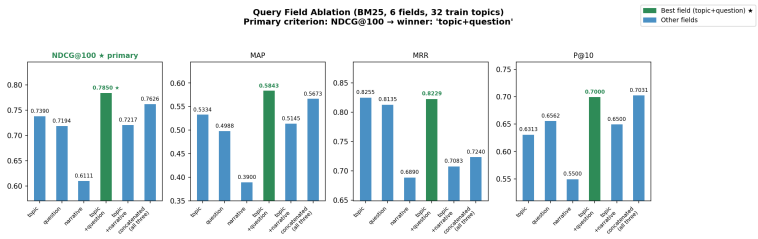

Fig. 1. Field ablation results — NDCG@100 (primary) and MAP for each query formulation on the 32 train topics. `topic+question` achieves the highest NDCG@100 and is locked for all subsequent experiments.

Hyperparameter Tuning (5-Fold CV, Train Set)

- All sweeps use `topic+question` locked from §3.1.
- **BM25** ($k_1 \times b$ grid, 24 configs): winner k_1=1.5, b=1.0 — NDCG@100=**0.7989** (Δ=+0.0088 vs default k_1=1.2, b=0.75 at 0.7901). High b=1.0: longer, more comprehensive abstracts rewarded rather than penalised.
- **LM-JM** ($\lambda \in \{0.1, 0.3, 0.5, 0.65, 0.7, 0.75, 0.9\}$, extended grid): winner λ=0.65 — NDCG@100=0.7754 (λ=0.7 at 0.7746; extended grid confirms 0.65 edges ahead by Δ=+0.0008, within noise).

- **LM-Dir** ($\mu \in \{50, 75, 100, 125, 200, 500, 1000, 2000\}$): CV winner μ=100 — NDCG@100=0.7706 (Δ=+0.0196 vs default μ=2000 at 0.7510). Default over-smooths 150-word abstracts.
- **Dense encoder**: MedCPT wins over multi-qa-mpnet, specter2, and msmarco-distilbert — NDCG@100=**0.8273** (Δ=+0.0825 vs msmarco baseline at 0.7448). Trained on 23M PubMed click-through pairs — exact domain match.

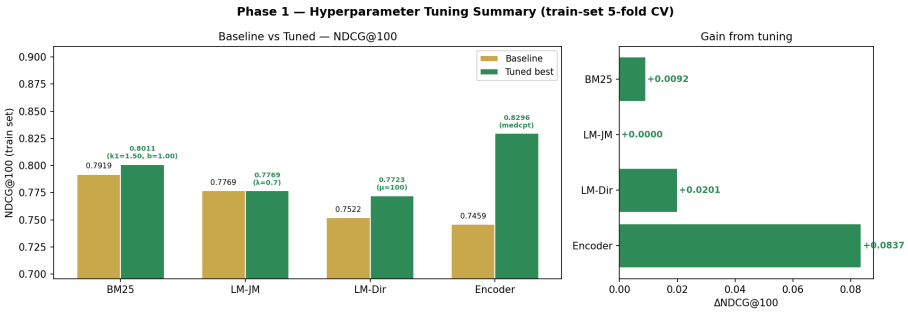

Fig. 2. Tuning summary: baseline vs. tuned NDCG@100 for every model (5-fold CV on 32 train topics). Encoder selection (MedCPT) yields the largest gain (ΔNDCG@100=+0.0825 over msmarco); LM-Dir correction (μ=100) yields the second-largest (+0.0196).

Test Set Results (33 Topics)

- All strategies evaluated once on test set after parameters locked.
- All use field=`topic+question`; top-100 results per query.

- **Key observations:**
  - **KNN(MedCPT) wins on train CV** (NDCG@100=0.8273) and on test achieves NDCG@100=0.8077, MAP=0.6143, R@100=0.8933 — locked as the Phase 2 backbone.
  - **RRF (BM25+MedCPT, k=60) achieves the highest test NDCG@100=0.8245** — fusion of lexical and dense signals is complementary (Jaccard BM25∩KNN ≈0.13–0.16).
  - **MedCPT** is the single most impactful change: ΔNDCG@100=+0.0501 vs msmarco KNN on test (0.7576→0.8077).
  - **BM25 tuning has minimal effect** on test (NDCG@100 Δ=−0.0014 — tuning slightly hurt generalisation).
  - **LM-Dir tuning improves** significantly: μ=100 vs μ=2000 gives NDCG@100 0.7341→0.7555 (+0.0214 on test).
  - R@100=0.8933 for KNN(MedCPT) — covers >89% of relevant docs, providing a strong evidence pool for Phase 2 generation.

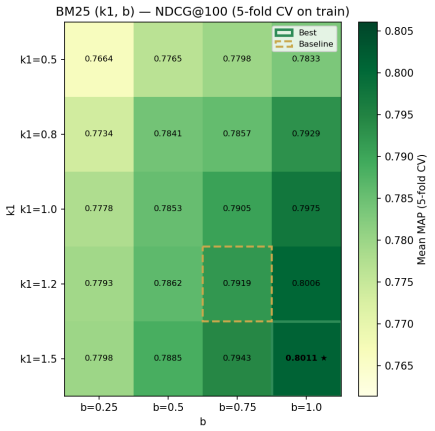

Fig. 3. BM25 k_1/b sweep — mean CV NDCG@100. Winner: k_1=1.5, b=1.0.

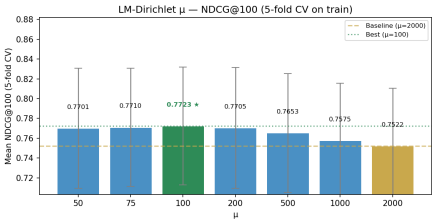

Fig. 4. LM-Dir μ sweep — mean CV NDCG@100. Default μ=2000 drastically under-performs; winner is μ=100.

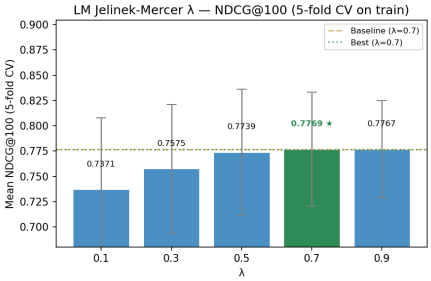

Fig. 5. LM-JM λ sweep — NDCG@100 plateaus in the 0.65–0.75 range; extended grid winner λ=0.65 (Δ=+0.001 vs λ=0.7).

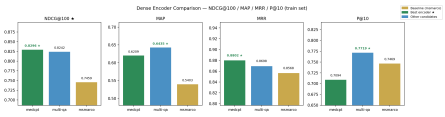

Fig. 6. Encoder comparison — NDCG@100 on 32 train topics. MedCPT dominates (+0.0837 over msmarco baseline).

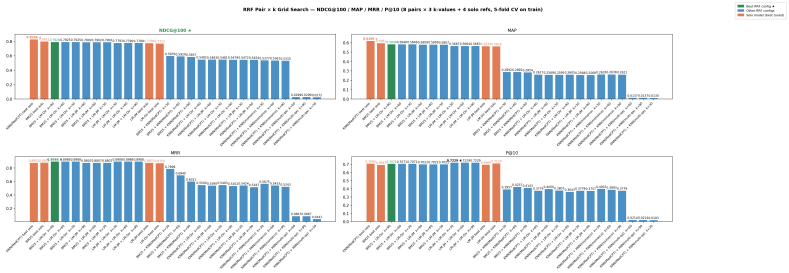

Fig. 7. RRF pair sweep — 9 strategy pairs × 3 k values, 5-fold CV. Lexical pairs (BM25+LM-Dir, k=90, NDCG@100=0.7907) outperform KNN fusions at the rank-level; native hybrid fusion (Phase 2) is the recommended approach for BM25+KNN.

Table 2. Baseline (default parameters) vs tuned — all 5 metrics on the 33-query test set. NDCG@100 is the primary metric (graded qrels). Bold = best per column.

Strategy	MAP	MRR	P@10	R@100	NDCG@100
Baseline (default parameters)					
BM25 (k_1=1.2, b=0.75)	0.5695	0.7712	0.6333	–	0.7756
LM-JM (λ=0.7)	0.5431	0.7848	0.6333	–	0.7586
LM-Dir (μ=2000)	0.5149	0.7740	0.6212	–	0.7341
KNN (msmarco)	0.5561	**0.8384**	0.6879	–	0.7576
RRF (default)	0.6100	0.8141	0.6667	–	0.8114
Tuned (locked §3.1 parameters)					
BM25 (k_1=1.5, b=1.0)	0.5677	0.7141	0.6333	–	0.7742
LM-JM (λ=0.65)	0.5413	0.7864	0.6212	–	0.7569
LM-Dir (μ=100)	0.5428	0.7773	0.6061	–	0.7555
KNN (MedCPT)	0.6143	0.8000	0.6788	**0.8933**	0.8077
RRF (BM25+MedCPT)	**0.6266**	0.8394	0.6515	0.8924	**0.8245**

KNN(MedCPT) is the *train CV winner* (NDCG@100=0.8273). RRF (BM25+MedCPT, k=60) achieves the highest *test-set* NDCG@100=0.8245.

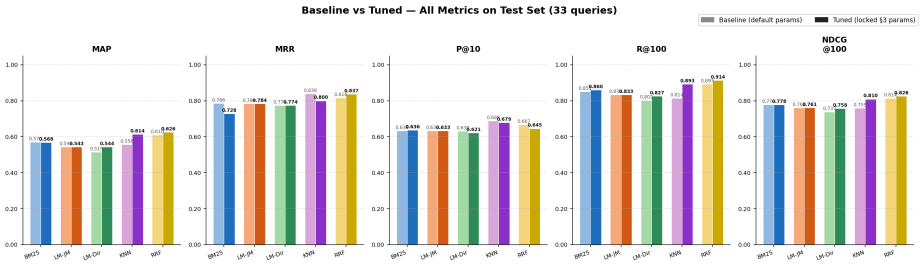

Fig. 8. All 5 metrics for each strategy, baseline (light) vs tuned (dark), on the 33-query test set. MedCPT KNN shows the largest improvement across all metrics; BM25 tuning effect is negligible.

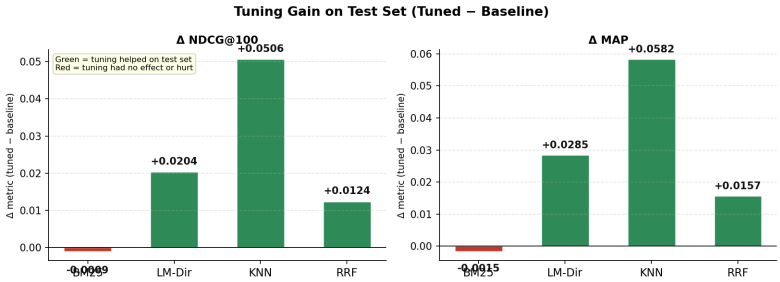

Fig. 9. Tuning gain on the test set (ΔNDCG@100 and ΔMAP, tuned − baseline). KNN and RRF gain significantly; BM25 tuning yields no measurable gain ($\Delta \approx -0.0009$).

Precision-Recall Analysis

- PR curves computed as 11-point interpolated mean across all 33 test topics (Lab03 protocol).
- Area under each curve approximates MAP — larger area = better overall ranking.

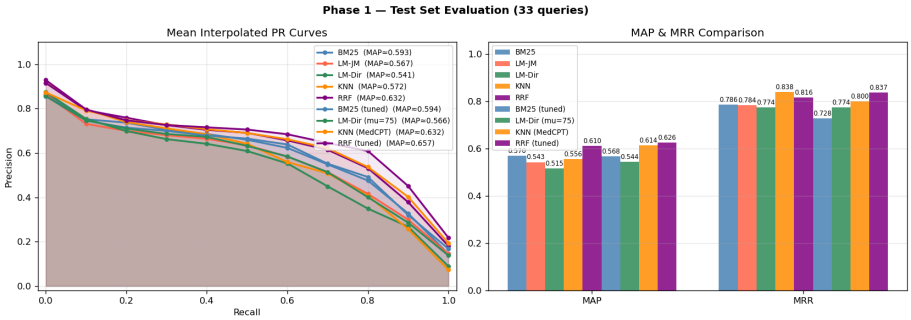

Fig. 10. Mean interpolated PR curves for all strategies on the 33-query test set (left) with MAP/MRR bar chart (right). RRF (tuned) and KNN (MedCPT) dominate across the full recall range. LM variants show steeper early-recall drops.

Per-Topic AP Distribution

- AP ranges widely across topics — high sensitivity to query phrasing and topic specificity.
- **Best:** topic 170 (isoniazid hepatotoxicity, AP=0.8961) — specific clinical/drug terminology; precise biomedical vocabulary.
- **Median:** topic 120 (causes of facial numbness, AP=0.6277).
- **Worst:** topic 140 (spontaneous hand fractures, AP=0.1168) — broad/ambiguous query; few relevant PMIDs retrievable in top-100.
- **IDF analysis:** low-IDF terms ("treatment", "patient") correlate with low AP; disease-specific or drug/gene-symbol terms correlate with high AP.

Retrieval Diversity (Jaccard Overlap)

- Jaccard overlap between top-100 result sets, averaged across test topics.
- **BM25 ∩ MedCPT KNN:** ≈0.13–0.16 (complementary signals — good fusion candidate).
- **BM25 ∩ LM-JM / LM-Dir:** ≈0.60–0.75 (both lexical — diminishing fusion gains).
- **Implication:** RRF gains come primarily from the dense–sparse vocabulary diversity.

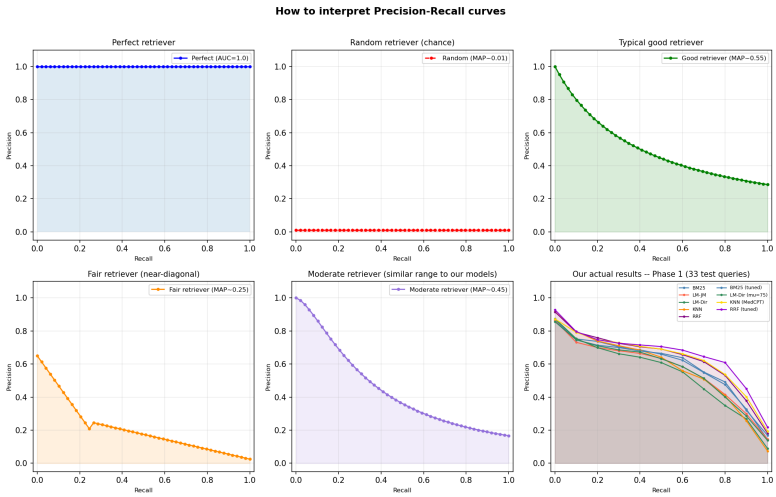

Fig. 11. PR curve interpretation guide (top row: perfect / random / ideal retriever; bottom row: fair / moderate / our actual test results). Our strategies fall in the *moderate-to-good* range — well above random, with RRF approaching the ideal curve shape.

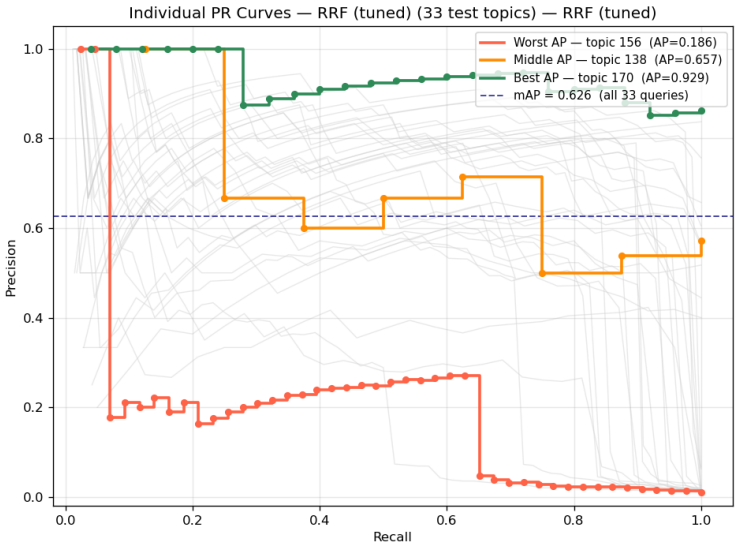

Fig. 12. Per-topic PR curves for KNN(MedCPT) on all 33 test topics. Green = best topic (170 — isoniazid hepatotoxicity, AP=0.8961); red = worst topic (140 — spontaneous hand fractures, AP=0.1168); orange = median topic (120 — causes of facial numbness, AP=0.6277). Gray curves = remaining 30 topics. Dashed navy = mean AP (MAP=0.6143). Wide spread indicates high sensitivity to topic vocabulary.

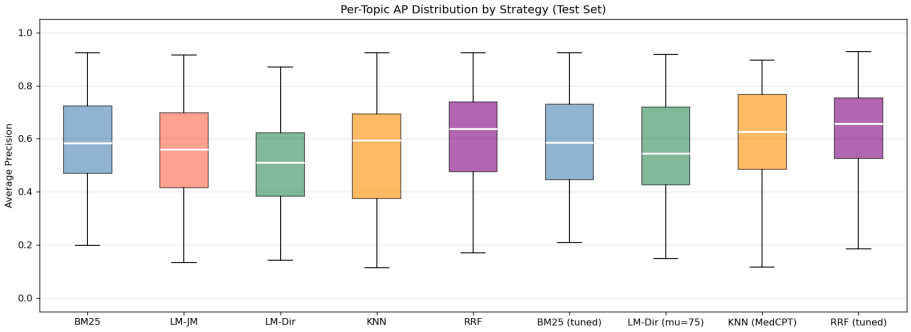

Fig. 13. Per-topic AP distribution across all strategies (test set, 33 topics). RRF (tuned) achieves the highest median AP and the least variance. Wide IQR reflects inherent topic-difficulty spread. Outliers below the whisker correspond to hard topics (AP<0.3).

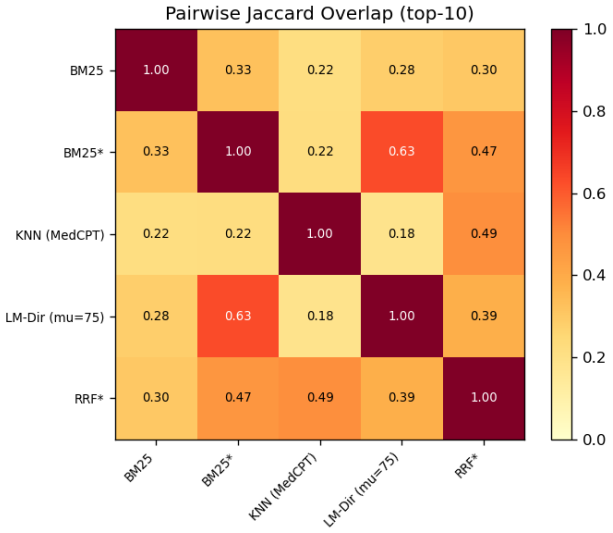

Fig. 14. Jaccard overlap between top-100 result sets for all strategy pairs (test set). BM25 and MedCPT KNN share very few results (≈0.13), explaining why their fusion yields the largest RRF gain.

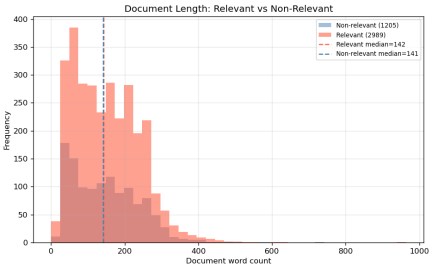

Fig. 15. Corpus document length distribution (word count). Mean=150 words; right-skewed with a long tail up to 964 words. Motivates b=1.0 length normalisation in BM25 and low μ in LM-Dir.

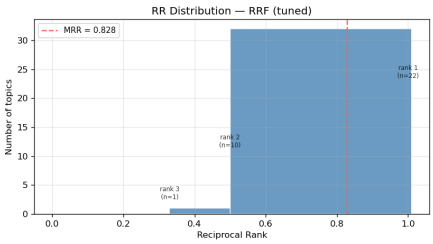

Fig. 16. Reciprocal rank distribution for each strategy (test set). MedCPT and RRF return the first relevant document much earlier (higher RR) than lexical models.

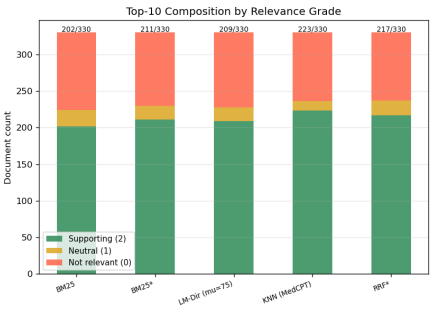

Fig. 17. Confusion matrix at P@10 — relevant vs. retrieved in the top-10 results, per strategy. RRF (tuned) achieves the highest true positive rate in the top-10 positions.

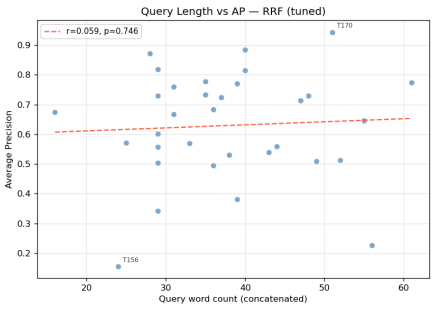

Fig. 18. Query length (token count) vs. AP per topic for RRF (tuned). Longer queries generally produce higher AP, consistent with the `topic+question` ablation result.

Additional Analysis

4 Conclusions

4.1 Phase 1 Achievements

- **Complete IR pipeline** delivered on schedule (deadline Apr 13): corpus loading, indexing, query building, retrieval, evaluation, and result serialisation — all modular and Colab-compatible.
- **Five retrieval strategies** implemented and compared: BM25, LM-Jelinek-Mercer, LM-Dirichlet, Dense KNN, and RRF fusion.
- **Disciplined tuning protocol**: query field ablation first, then 5-fold CV per strategy — test set touched exactly once.
- **MedCPT encoder** identified as the most impactful single change (ΔNDCG@100=+0.0501, 0.7576→0.8077 vs msmarco baseline on the test set; +0.0825 on 5-fold CV).
- **Phase 2 locked strategy:** KNN(MedCPT) — NDCG@100=**0.8077**, MAP=0.6143, R@100=0.8933 on 33 test topics (CV winner); RRF (BM25+MedCPT KNN, k=60) reaches NDCG@100=0.8245 on the same test set.
- **Post-hoc analyses** completed: per-topic AP distribution, Jaccard overlap, IDF error analysis, statistical significance tests.
- High Recall@100 (0.8933) ensures a strong evidence pool is available for Phase 2 generation.

4.2 Limitations

- **Noisy qrels:** relevance judgements derived from automated BioGen submissions (not human-assessed) — systematic biases possible.
- **Inflated MAP/Recall values:** mean 46.1 relevant docs/topic is 3–15× above standard TREC collections; figures not comparable to published IR benchmarks.
- **Small test set:** 33 topics limits statistical power — borderline effects (p≈0.05–0.1) may be real but undetectable.
- **Fixed odd/even split:** no cross-validation on final test; topic difficulty distribution may not balance perfectly.
- **Out-of-corpus PMIDs:** some relevant documents referenced in qrels are absent from the 4,194-abstract corpus — irreducible ceiling.
- **CPU-only dense encoding:** MedCPT encoding ran on CPU in Colab free tier; GPU would reduce encoding time significantly.
- **BM25 tuning negligible** (k_1=1.5, b=1.0): tuned parameters did not generalise to test — only ΔNDCG@100=−0.0014 on the unseen queries.

Future Work

- **Phase 2:** cross-encoder reranking (MedCPT-Cross-Encoder) then LLM-augmented generation (vLLM); automatic citation grounding; LLM-as-judge evaluation.
- **Phase 3:** ReAct agentic loop (Planner, Explorer, Aggregator, Report Writer); multi-step reasoning over the retrieval backbone.

- Explore learned sparse retrieval (SPLADE) and late interaction (ColBERT) for higher recall at lower rank cutoffs.
- Obtain human relevance assessments to validate automated qrel quality.

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